

<i>Please ensure this tracking sheet is completed as a group, as it will be used to monitor your individual participation within the team.</i>						
Team Members	Role/ Responsibility	Email				
Jotham JABO RUTIJANA	Data cleaning, standardization, dynamic component selection, loadings interpretation, report writing	j.rutijana@alustudent.com				
Elie Kuradusenge	Data loading, PCA implementation (covariance, eigendecomposition, projection), visualization, Task 3 benchmarking	e.kuradusen@alustudent.com				
Task Allocation and Tracker						
	Task Allocated	Assigned Member	Deadline	Completion Status (Not Started/In Progress/Completed/Did not participate)	Reviewed by Team? (Yes/No)	Comments
	Dataset selection and loading (Kaggle African Crises) with NumPy genfromtxt	Elie Kuradusenge	25/9/2026	Completed	Yes	1059 rows × 14 columns, no Pandas used
	Data cleaning: identify numeric vs non-numeric columns, drop cc3, country, banking_crisis, impute missing values with median	Jotham JABO RUTIJANA	26/9/2026	Completed	Yes	NumPy-only imputation
	Exclude identifier columns (year, case, systemic_crisis) and standardize using z-score formula	Jotham JABO RUTIJANA	26/9/2026	Completed	Yes	Prevents PCA contamination from IDs
	Covariance matrix computation and eigendecomposition using np.linalg.eigh	Elie Kuradusenge	26/9/2026	Completed	Yes	Verified against np.cov; symmetric solver used
	Sort eigenvalues/eigenvectors descending and plot scree + cumulative variance	Jotham JABO RUTIJANA	26/9/2026	Completed	Yes	Visualized 90% variance threshold

	Dynamic component selection (np.searchsorted) and projection onto principal components	Elie Kuradusenge	28/9/2026	Completed	Yes	Kept 9 of 11 components at 93.29% variance
	Loadings interpretation for each PC and before/after PCA visualization	Jotham JABO RUTIJANA	28/9/2026	Completed	Yes	Identified driving features per PC
	Task 3: Naive vs fast PCA benchmarking and chunked covariance for large datasets	Elie Kuradusenge	28/9/2026	Completed	Yes	Demonstrated speedup of eigh over eig
	Written answers, GitHub repository setup, and combined PDF submission	Jotham JABO RUTIJANA	30/9/2026	Completed	Yes	Notebook, CSV, task sheet uploaded

Meeting Attendance Log

Meeting Date	Agenda	Facilitator	Attendees	Key Discussion Points	Next Steps	
24/9/2026		Elie Kuradusenge	Elie Kuradusenge Jotham JABO RUTIJANA	Reviewed PCA requirements, dataset requirements, and submission process	Complete implementation and upload to GitHub	
30/9/2026		Jotham JABO RUTIJANA	Elie Kuradusenge Jotham JABO RUTIJANA	Final executions, completed implementation and assignment submission	Marks	